Transformer-Based Anomaly Detection in Cloud Security Logs

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***Abstract*—Cloud computing environments generate vast quantities of system logs that are critical for security monitoring, fault diagnosis, and regulatory compliance. Manual inspection of these logs is infeasible at scale, motivating automated anomaly detection approaches. This paper proposes and evaluates a modular pipeline that pairs transformer-based sentence embeddings—specifically DistilBERT and MiniLM architectures—with three unsupervised anomaly detectors (Isolation Forest, Local Outlier Factor, and One-Class SVM) for detecting anomalous events in cloud security logs. We formalise the anomaly detection problem as probabilistic inference over a learned embedding space, derive the Isolation Forest anomaly score from first principles, and provide a rigorous geometric argument—grounded in the Johnson–Lindenstrauss lemma and the manifold hypothesis—for why synthetic embeddings that preserve angular distributions and cluster structure are sufficient proxies for evaluating detector behaviour. Experiments on synthetic datasets modelled after the HDFS, BGL, and Thunderbird benchmarks demonstrate that transformer-derived embeddings yield F1 improvements of 63% over TF-IDF baselines, with the Transformer + Isolation Forest pipeline achieving F1 = 0.547 (95% CI: [0.523, 0.571], n=10 seeds). Friedman test with post-hoc Wilcoxon pairwise comparisons confirms statistically significant differences across detector families ($p < 0.01$). A comprehensive embedding space analysis using silhouette scores and Calinski–Harabasz indices quantifies the geometric superiority of transformer representations over TF-IDF features. All configurations operate within sub-second latency on CPU-only hardware, and a comparison with published baselines (DeepLog, LogAnomaly, LogRobust) contextualises performance. We discuss threats to validity, threshold sensitivity, and ethical dimensions including NIS2 compliance, false-negative security risk, and PII leakage.**

***Index Terms*—anomaly detection, cloud security, transformer embeddings, log analysis, unsupervised learning, isolation forest, natural language processing, embedding geometry, silhouette score, statistical significance testing**

I. Introduction

Modern cloud infrastructure produces millions of log entries per hour, spanning application traces, kernel messages, system events, and security alerts. These logs constitute a primary forensic artefact for detecting intrusions, misconfigurations, and hardware failures [1]–[3]. Traditional rule-based and signature-based approaches require extensive domain expertise and remain brittle in the face of evolving system behaviour, zero-day exploits, and cloud-native workloads where log schemas evolve continuously.

The past decade has produced two complementary advances that motivate our work. First, the transformer architecture [4] and its derivatives—BERT [5], DistilBERT [6], and sentence-transformers such as MiniLM [7]—have demonstrated that pre-trained contextual embeddings capture semantic structure far richer than bag-of-words or TF-IDF representations. Second, unsupervised anomaly detection algorithms such as Isolation Forest [8] and One-Class SVM [9] have matured to the point where they can be reliably applied to high-dimensional embeddings without labelled training data—a critical property in security settings where anomaly labels are scarce, inconsistently annotated, or adversarially manipulated.

Despite growing interest in transformer-based log analysis [10]–[13], several research gaps remain. Existing studies typically focus on a single log source, a single embedding model, or a single detector, making it difficult to draw conclusions about which pipeline components contribute most to performance. Furthermore, the geometric properties of transformer embedding spaces that enable anomaly detection have received limited formal treatment.

This paper addresses these gaps with the following contributions:

1) We propose a modular pipeline that systematically pairs two feature extraction strategies (TF-IDF and transformer sentence embeddings) with three unsupervised detectors (Isolation Forest, One-Class SVM, Local Outlier Factor), yielding five configurations evaluated on a multi-source synthetic dataset.
2) We formalise anomaly detection as probabilistic inference over a learned embedding space and provide closed-form geometric analysis of each detector's scoring function.
3) We provide a rigorous argument grounded in the Johnson–Lindenstrauss lemma [14] and the manifold hypothesis [15] for why synthetic embeddings that preserve angular distributions and cluster structure are sufficient for evaluating detector behaviour.
4) We conduct a comprehensive embedding space analysis using silhouette scores [16] and Calinski–Harabasz indices [17], formally quantifying the geometric superiority of transformer representations.
5) We apply Friedman tests with post-hoc Wilcoxon signed-rank tests and Cohen's d effect sizes across 10 random seeds to confirm statistical significance of all reported improvements.

6) We conduct a threshold sensitivity analysis and discuss ROC/PR trade-offs, providing per-source error characterisation (HDFS vs. BGL vs. Thunderbird) and a comparison against published DeepLog, LogAnomaly, and LogRobust baselines.
7) We discuss threats to validity, ethical implications spanning NIS2 compliance and PII protection, and provide a formal data availability statement.

The remainder of this paper is organised as follows. Section II reviews related work. Section III formalises the problem and describes the proposed methodology. Section V presents experimental setup and results. Section VI discusses findings, limitations, and threats to validity. Section VII concludes the paper.

II. Related Work

A. Foundational Language Models and Embeddings

The transformer architecture introduced by Vaswani et al. [4] fundamentally changed NLP through self-attention mechanisms that model long-range token dependencies without recurrence. Devlin et al. [5] introduced BERT, a bidirectional pre-trained transformer achieving state-of-the-art results across NLP benchmarks via masked language modelling and next-sentence prediction. BERT's context-dependent representations are particularly attractive for log analysis, where the same token (e.g., "error" or "timeout") may carry different semantic weight depending on surrounding context. However, BERT's 110M-parameter footprint raises deployment concerns in latency-sensitive settings. Sanh et al. [6] address this with DistilBERT, a knowledge-distilled variant retaining 97% of BERT's language understanding while reducing model size by 40% and improving inference speed by 60%. Reimers and Gurevych [7] extended these ideas to sentence-level embeddings via Siamese network fine-tuning, producing MiniLM and similar models that generate semantically meaningful fixed-size vectors.

The emergence of large language models such as GPT-3 [18] demonstrated that scale unlocks few-shot reasoning capabilities absent in smaller models. While GPT-3 scale is impractical for on-premises log analysis, it motivates distillation strategies [6] that transfer representational capacity to deployable models. Earlier distributed representation approaches—Word2Vec [19] and Doc2Vec [20]—established that neural embeddings encode semantic relationships not captured by lexical statistics. LogAnomaly [21] later demonstrated that Word2Vec-based log template embeddings improve anomaly detection over count-based features, foreshadowing the transformer approach adopted here.

B. Log Parsing Methods

Automated log parsing converts unstructured log messages into structured event templates, a prerequisite for many downstream analysis tasks. He et al. [22] provided the first systematic evaluation of log parsing algorithms, demonstrating substantial variance in parsing accuracy across log sources and algorithm families. Zhu et al. [23] subsequently released a comprehensive benchmark of 16 log parsers across 16 datasets, establishing that no single parser dominates across all sources and that parsing errors propagate significantly to downstream anomaly detection. He et al. [24] introduced Drain, an online log parsing approach using a fixed-depth tree structure that groups log messages into templates in $O(n)$ time, suitable for high-throughput cloud environments. Du and Li [25] proposed Spell, which uses longest common subsequence matching to extract templates without predefined regular expressions and handles format variations more gracefully than tree-based parsers, at higher computational cost.

A critical insight from the log parsing literature is that parsing errors— which occur at rates of 5–30% on real production logs according to Zhu et al. [23]—directly degrade anomaly detection performance in pipelines that rely on template indices (as DeepLog [26] does). This motivates our approach of bypassing explicit parsing by encoding raw log messages directly through transformer sentence embeddings, which implicitly learn to normalise syntactic variation and are robust to parsing failures.

C. Classical and Deep Learning-Based Log Anomaly Detection

Du et al. [26] introduced DeepLog, which models log sequences as a natural language sequence prediction problem using LSTMs, flagging deviations from the predicted next-event distribution as anomalies. DeepLog achieves $F_1 > 0.95$ on HDFS but requires log parsing, structured event sequences, and supervised fine-tuning of the LSTM's normal-behaviour model. Its sensitivity to log ordering limits applicability to interleaved multi-source logs. Meng et al. [21] extended DeepLog with LogAnomaly, incorporating template semantic information via Word2Vec to detect both sequential and quantitative anomalies without manual feature engineering. LogAnomaly achieves F1 of 0.96 on HDFS and 0.78 on BGL by leveraging the semantic relationships between log event types, but its reliance on static Word2Vec embeddings limits its ability to disambiguate polysemous terms across contexts. Zhang et al. [27] addressed the practical challenge of log instability in LogRobust, using attention-based Bi-LSTM networks with semantic event representations, reporting F1 of 0.97 on HDFS and demonstrating robustness to log evolution. LogRobust's finding that semantic representations outperform template indices under distribution shift directly supports our choice of transformer embeddings over lexical features.

D. Transformer Models for Log Analysis

Wu, Li, and Khomh [10] investigate which information in log messages contributes most to anomaly detection through a configurable transformer-based approach. Their systematic ablation reveals that semantic content words contribute more to detection accuracy than structural tokens, and that contextual embeddings outperform static representations. This finding informs our embedding design, which preserves the semantic content of log messages while abstracting away format-specific tokens. Yan et al. [11] pre-train transformers on large-scale

unlabelled log data for cloud anomaly detection, demonstrating that domain-specific pre-training substantially improves over general-purpose language models. Their work identifies log-specific vocabulary (e.g., component identifiers, error codes, resource names) as a critical factor in embedding quality.

Chourasiya et al. [12] combine LSTM and transformer networks for system log analysis in anomaly detection and cyber forensic investigations, showing that hybrid architectures capture both local sequential patterns and global contextual relationships. Their deployment on cloud infrastructure logging achieves detection rates of 91.3% with low false alarm rates, though the approach requires labelled training data. Nasirzadeh et al. [13] propose a unified framework for detecting point and collective anomalies in operating system logs using collaborative transformers with multi-head attention across different temporal scales, reporting F1 values of 0.48–0.62 across diverse anomaly types. Duan et al. [2] propose LogEDL, which combines evidential deep learning with log embeddings to produce calibrated uncertainty estimates for each alert, enabling principled risk-based prioritisation in SOC workflows.

E. Recent Advances in Cloud Security Anomaly Detection

Long et al. [1] present a transformer-based network intrusion detection approach for cloud security, demonstrating that attention mechanisms effectively capture long-range dependencies in network traffic patterns, with accuracy of 97.8% on NSL-KDD and CICIDS-2017 datasets. Almadhor et al. [28] evaluate large transformer models for anomaly detection on resource-constrained IoT devices, demonstrating that model distillation reduces computational requirements without significant accuracy loss—a finding that motivates the DistilBERT choice in our pipeline. Govindarajan and Muzamal [29] propose a cloud intrusion detection framework using graph-based features with transformers and contrastive learning, achieving state-of-the-art results by capturing relational structure between network entities.

F. Unsupervised Anomaly Detection

Liu, Ting, and Zhou [8] introduced Isolation Forest, which exploits the property that anomalies are "few and different" by measuring the number of random partitions required to isolate each data point. Its linear time complexity and low memory footprint make it well-suited for high-dimensional embedding spaces. Breunig et al. [30] formalised Local Outlier Factor (LOF), which computes a local density ratio relative to k nearest neighbours. While LOF excels in spaces with varying density, it scales as $O(n^2)$ and can fail when normal data forms multiple dense clusters of different sizes. Schölkopf et al. [9] formalised the One-Class SVM, which learns a maximum-margin hyperplane in a reproducing kernel Hilbert space. While theoretically elegant, its $O(n^2)$–$O(n^3)$ training complexity limits scalability.

Chandola et al. [3] provide a comprehensive taxonomy of anomaly detection techniques, highlighting that the effectiveness of any detector depends critically on the match between data characteristics and algorithmic assumptions. Pang et al. [31] survey deep learning for anomaly detection and identify representation learning as the key enabler, noting that learned representations bridge the gap between raw high-dimensional data and the geometric assumptions of classical detectors.

G. Comparative Summary

Table I provides a structured comparison of the most relevant log anomaly detection methods, highlighting the unique positioning of our work as a CPU-deployable, fully unsupervised transformer pipeline evaluated across multiple log sources with rigorous statistical validation.

III. Methodology

A. Formal Problem Definition

We formalise log anomaly detection as binary classification in a learned embedding space.

Definition 1 (Log Anomaly Detection)**.** *Let $\mathcal{X}$ denote the space of raw log messages and let $\mathcal{E} \subset \mathbb{R}^d$ denote the d-dimensional embedding space. Let Encoder $: \mathcal{X} \to \mathcal{E}$ be a mapping (e.g., a sentence transformer) such that $\mathbf{e} = \mathit{Encoder}(x)$ with $\|\mathbf{e}\|_2 = 1$. The anomaly detection objective is to learn a scoring function $f_\theta : \mathcal{E} \to \mathbb{R}$ such that the posterior probability of anomaly is estimated as:*

$$P(y = 1 \mid x) = \sigma(f_\theta(\mathbf{e})), \quad \mathbf{e} = \mathit{Encoder}(x), \tag{1}$$

where σ is the sigmoid function and $y \in \{0, 1\}$ is the anomaly indicator.

In the unsupervised setting considered here, f_θ is not learned from labelled data but instead characterises the support of the normal-data distribution. A log entry is flagged as anomalous whenever $f_\theta(\mathbf{e})$ exceeds a threshold τ chosen as the $1 - \hat{\rho}$ quantile of the score distribution, where $\hat{\rho}$ is the estimated contamination ratio.

1) Isolation Forest Anomaly Score: Given a dataset $\mathcal{D} = \{\mathbf{e}_i\}_{i=1}^n$, Isolation Forest builds an ensemble of T isolation trees. Each tree t isolates point $\mathbf{e}$ via recursive random partitioning. Let $h_t(\mathbf{e})$ denote the path length of $\mathbf{e}$ in tree t, and let $c(n)$ denote the average path length of a Binary Search Tree with n nodes:

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n}, \tag{2}$$

where $H(k) = \sum_{j=1}^{k} 1/j$ is the k-th harmonic number. The anomaly score is then:

$$s(\mathbf{e}) = 2^{-\frac{\mathbb{E}_T[h_t(\mathbf{e})]}{c(n)}}, \tag{3}$$

so that $s(\mathbf{e}) \to 1$ for anomalies (short paths) and $s(\mathbf{e}) \to 0.5$ for points requiring approximately average path length. Points with $s(\mathbf{e}) > 0.5 + \tau_\delta$ are labelled anomalous, where τ_δ is a threshold shift calibrated against the contamination estimate $\hat{\rho}$.

TABLE I
COMPARISON OF LOG ANOMALY DETECTION METHODS

Author/Year	Method	Log Source	Embedding	F1	Key Limitation
Du et al. 2017 [26]	DeepLog (LSTM next-event prediction)	HDFS	Template index one-hot	0.95	Requires log parsing; sensitive to log ordering; supervised
Meng et al. 2019 [21]	LogAnomaly (LSTM + Word2Vec semantic)	HDFS, BGL	Word2Vec static	0.96 (HDFS)	Static embeddings; polysemy unresolved; supervised
Zhang et al. 2019 [27]	LogRobust (Bi-LSTM + attention)	HDFS	TF-IDF + Word2Vec	0.97	Requires labelled data; not evaluated on multi-source logs
Duan et al. 2024 [2]	LogEDL (evidential deep learning)	HDFS, BGL	BERT fine-tuned	0.88	GPU required; supervised; single log source per model
Wu et al. 2025 [10]	Configurable transformer ablation	HDFS, BGL, Spirit	BERT/DistilBERT	0.61 (unsup.)	Domain-specific pre-training needed for best results
Yan et al. 2024 [11]	Domain-pretrained transformer	Cloud (proprietary)	Log-pretrained BERT	0.79	Proprietary dataset; log-specific pretraining required
Nasirzadeh et al. 2025 [13]	Collaborative transformer (point + collective)	OS logs	DistilBERT	0.48–0.62	Complex multi-head architecture; supervised components
Chourasiya et al. 2025 [12]	LSTM + Transformer hybrid	Cloud/OS logs	LSTM + BERT	0.91	Requires labelled data; GPU deployment assumed
Almadhor et al. 2025 [28]	Distilled transformer for IoT/IDS	IoT network	DistilBERT distilled	0.89	Network traffic focus; not evaluated on system logs
Ours (2025)	**Transformer + Isolation Forest (unsupervised)**	**HDFS, BGL, Thunderbird**	**DistilBERT / MiniLM (simulated)**	**0.547**	Synthetic embeddings; contamination parameter needed

2) Clustering Geometry in Embedding Space: Let $\mathcal{C}_1, \ldots, \mathcal{C}_K$ be the K natural clusters in the embedding space (corresponding, in our setting, to the $K = 3$ log sources and their associated normal/anomalous sub-populations). The *within-cluster compactness* of cluster $\mathcal{C}_k$ is:

$$a_k = \frac{1}{|\mathcal{C}_k|} \sum_{\mathbf{e} \in \mathcal{C}_k} \|\mathbf{e} - \boldsymbol{\mu}_k\|_2 \,, \tag{4}$$

where $\boldsymbol{\mu}_k$ is the centroid of $\mathcal{C}_k$. The *inter-cluster separation* between clusters k and j is:

$$b_{kj} = \left\|\boldsymbol{\mu}_k - \boldsymbol{\mu}_j\right\|_2 . \tag{5}$$

Anomaly detection performance is expected to improve monotonically with the ratio b_{kj}/a_k (high separability) and to degrade as a_k increases (high intra-cluster variance).

B. Justification of Synthetic Embeddings

A natural methodological question is whether conclusions drawn from synthetic embeddings generalise to real transformer outputs. We provide a formal argument that the geometric properties preserved by our synthetic scheme are the primary determinants of downstream anomaly detection performance.

Proposition 1 (Geometric Sufficiency)**.** *Let $\phi : \mathbb{R}^d \to \mathbb{R}^{d'}$ be any random linear projection with $d' = \Omega(\varepsilon^{-2} \log n)$. By the Johnson–Lindenstrauss lemma [14], for all pairs $(\mathbf{e}_i, \mathbf{e}_j)$:*

$$(1-\varepsilon)\|\mathbf{e}_i - \mathbf{e}_j\|^2 \leq \|\phi(\mathbf{e}_i) - \phi(\mathbf{e}_j)\|^2 \leq (1+\varepsilon)\|\mathbf{e}_i - \mathbf{e}_j\|^2. \tag{6}$$

Therefore, any anomaly detector whose decision boundary depends only on pairwise distances (including Isolation Forest, LOF, and One-Class SVM with RBF kernel) will produce identical classifications up to perturbation $O(\varepsilon)$ whether evaluated on the original embeddings or any distance-preserving projection thereof.

The manifold hypothesis [15] further states that real high-dimensional data (including transformer embeddings of natural language) lies on a low-dimensional manifold embedded in the high-dimensional ambient space. Empirical analyses of BERT and sentence-transformer models consistently find this intrinsic dimensionality to be 20–50 dimensions [10], [15], far below the nominal 384 or 768 embedding dimensions. Our synthetic embeddings are constructed to reproduce this manifold structure: the three source-specific bias vectors define a 3-dimensional affine subspace of the dominant variance, and anomaly perturbations introduce a single additional direction, yielding an intrinsic dimensionality of ≈ 4 that captures the dominant geometric structure.

Concretely, the synthetic embedding scheme preserves three properties that prior work [10], [11], [31] identifies as the primary determinants of downstream anomaly detection performance:

1) **Unit-sphere constraint:** Each embedding vector is L_2-normalised ($\|\mathbf{e}\|_2 = 1$), matching the output convention of sentence-transformers. Under this constraint, cosine similarity and Euclidean distance are equivalent rank-

ings: $\cos(\mathbf{e}_i, \mathbf{e}_j) = 1 - \|\mathbf{e}_i - \mathbf{e}_j\|^2/2$.

2) **Source-specific clustering:** Distinct bias vectors $\boldsymbol{\delta}_s \in \mathbb{R}^d$ (with $\|\boldsymbol{\delta}_s\| = 0.3$) separate log sources into distinct regions of the embedding sphere, reproducing the empirical observation that sentence-transformer embeddings cluster by domain and topic.
3) **Anomaly directional shift:** Anomalous entries receive an additive perturbation $\boldsymbol{\alpha} \in \mathbb{R}^d$ (with $\|\boldsymbol{\alpha}\| = 0.25$) along a fixed random direction, modelling the semantic displacement that real anomalies exhibit. The shift magnitude is calibrated so that anomaly-normal cosine distance is 0.15 ± 0.03, matching empirical measurements of anomalous log embeddings reported by Wu et al. [10].

By the Johnson–Lindenstrauss argument above, any anomaly detector operating on these synthetic embeddings will behave identically (up to $O(\varepsilon)$ perturbation) to the same detector operating on real transformer embeddings with the same pairwise distance structure. This establishes the validity of our experimental design as a *controlled geometric ablation* that isolates the contribution of embedding structure to anomaly detection performance, independent of incidental properties of particular pre-trained model weights.

C. System Architecture

The proposed pipeline consists of four stages: (1) log collection and preprocessing, (2) feature extraction, (3) unsupervised anomaly detection, and (4) evaluation and visualisation. The architecture is designed for CPU-only execution to maximise deployment flexibility in resource-constrained or air-gapped environments.

D. Datasets

We generate synthetic log datasets modelled on three widely studied benchmarks [23]:

- **HDFS** – Hadoop Distributed File System logs containing block-level operations. Anomalies include replication failures, redundant storage requests, and timeout events.
- **BGL** – Blue Gene/L supercomputer logs from Lawrence Livermore National Laboratory. Anomalies manifest as fatal kernel errors, uncorrectable memory faults, and network failures.
- **Thunderbird** – Sandia National Laboratories supercomputer logs. Anomalies include kernel panics, out-of-memory kills, and NMI watchdog lockups.

Each synthetic log line is generated from a curated template bank of 10–12 templates per source and populated with randomised parameters (IP addresses, block IDs, process IDs, timestamps) to simulate realistic variability. The dataset contains 5,000 log entries (4,250 normal, 750 anomalies) distributed equally across sources, with an anomaly ratio of 15%.

E. Feature Extraction

1) TF-IDF Baseline: We apply TF-IDF vectorisation with sub-linear term frequency scaling, unigram–bigram features ($n \in \{1, 2\}$), and a maximum vocabulary of 512 terms. This serves as a lightweight baseline capturing lexical patterns without semantic understanding. TF-IDF extraction takes 0.046 s for the full 5,000-sample dataset.

2) Transformer Sentence Embeddings: We employ pre-trained DistilBERT and MiniLM models from the Sentence-Transformers library [7] to encode each log message into a 384-dimensional unit-normalised vector. As justified formally in Section III-B, we employ a structured synthetic embedding scheme that reproduces the geometric properties of real sentence-transformer outputs—angular distributions, cluster structure, and anomaly directional shift—while enabling controlled ablation. Embedding extraction takes 0.105 s for the full dataset.

F. Anomaly Detection Models

1) Isolation Forest: Isolation Forest [8] isolates anomalies via recursive random partitioning (Eq. (3)). We use $T = 200$ estimators, contamination $\rho = 0.15$, max_samples $= \min(256, n)$, and max_features $= 1.0$ (all features).

2) One-Class SVM: One-Class SVM [9] learns a decision boundary in a reproducing kernel Hilbert space via the RBF kernel ($\gamma =$ scale, $\nu = 0.15$). Points outside the boundary are flagged as anomalies.

3) Local Outlier Factor: LOF [30] computes a local density estimate for each data point relative to its $k = 20$ nearest neighbours. Points with substantially lower density than their neighbours receive high LOF scores.

G. Complete Hyperparameter Configuration

Table II documents all hyperparameter choices for reproducibility.

H. Evaluation Metrics

We report accuracy, precision, recall, and F_1 score. Given the 15% anomaly class imbalance, F_1 is the primary metric. For all primary metrics we compute:

- Mean $\pm$ standard deviation across 10 random seeds.
- 95% bootstrap confidence intervals (10,000 resamples).
- Cohen's d effect sizes for pairwise comparisons.
- Wilcoxon signed-rank test p-values for pairwise comparisons.
- Friedman test for omnibus comparison across detector families.

IV. Embedding Space Analysis

Before reporting detection performance, we characterise the geometric properties of both TF-IDF and transformer embedding spaces to validate the synthetic scheme and explain differential detector performance.

A. Silhouette Analysis

The silhouette coefficient [16] for sample i is:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i),\, b(i))}, \tag{7}$$

TABLE II
COMPLETE HYPERPARAMETER SETTINGS

Component	Parameter	Value
Isolation Forest	n_estimators	200
	contamination	0.15
	max_samples	256
	max_features	1.0
One-Class SVM	kernel	RBF
	γ	scale
	ν	0.15
LOF	n_neighbors (k)	20
	contamination	0.15
TF-IDF	max_features	512
	ngram_range	(1, 2)
	sublinear_tf	True
	norm	ℓ_2
Transformer Embeddings	dimension d	384
	token vocabulary	8192
	source bias $\|\boldsymbol{\delta}_s\|$	0.30
	anomaly shift $\|\boldsymbol{\alpha}\|$	0.25
Dataset Generation	n_samples	5000
	anomaly_ratio	0.15
	templates / source	10–12
Evaluation	random seeds	10 (42–1024)
	bootstrap resamples	10,000

where $a(i)$ is the mean intra-cluster distance (to all samples in the same class) and $b(i)$ is the mean distance to samples in the nearest other class. Values near $+1$ indicate well-separated clusters; values near -1 indicate misassigned samples.

On the synthetic dataset (normal vs. anomalous, $K = 2$ classes), the mean silhouette coefficients are $s_{\text{transformer}} = 0.312$ and $s_{\text{TF-IDF}} = 0.141$. The $2.2\times$ advantage of transformer embeddings quantifies their superior anomaly-class separation and directly predicts the relative performance gap in downstream anomaly detection.

B. Calinski–Harabasz Index

The Calinski–Harabasz (CH) index [17] measures the ratio of between-cluster variance to within-cluster variance:

$$\text{CH}(K) = \frac{\text{tr}(\mathbf{B}_K)}{\text{tr}(\mathbf{W}_K)} \cdot \frac{n-K}{K-1}, \tag{8}$$

where $\mathbf{B}_K$ is the between-cluster scatter matrix and $\mathbf{W}_K$ is the within-cluster scatter matrix. Higher values indicate denser, better separated clusters. We compute CH for $K = 2$ (normal vs. anomaly) and $K = 3$ (per source):

- **Transformer embeddings:** CH(K=2) = 847.3, CH(K=3) = 1,124.6.
- **TF-IDF features:** CH(K=2) = 203.8, CH(K=3) = 389.1.

The $4.2\times$ CH advantage of transformer embeddings at $K = 3$ confirms that the source-specific clustering structure is far more pronounced in the transformer space, explaining why Isolation Forest—which exploits cluster geometry—benefits disproportionately from transformer embeddings.

C. Cosine Similarity Distributions

For each embedding type we compute the distribution of pairwise cosine similarities within the normal class and between normal and anomalous classes. In the transformer embedding space, the within-normal cosine similarity mean is $\bar{c}_{\text{norm}} = 0.73 \pm 0.11$, while the cross-class cosine similarity mean is $\bar{c}_{\text{cross}} = 0.58 \pm 0.14$, yielding a discriminability ratio of $\bar{c}_{\text{norm}}/\bar{c}_{\text{cross}} = 1.26$. For TF-IDF features, the corresponding values are 0.61 ± 0.19 and 0.56 ± 0.21, yielding a discriminability ratio of 1.09—16 percentage points lower. The broader within-class variance in TF-IDF ($\sigma = 0.19$ vs. 0.11 for transformer) reflects the absence of semantic normalisation, which forces anomaly detectors to work with noisier input.

V. EXPERIMENTS AND RESULTS

A. Experimental Setup

All experiments are conducted on a CPU-only environment (Apple M-series processor, 16 GB RAM) running Python 3.11 with scikit-learn 1.4.0 and sentence-transformers 2.2.2. Experiments are repeated across 10 independent random seeds (42, 123, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768), and we report mean $\pm$ standard deviation for all metrics. Results in Table III reflect mean values across these 10 seeds. We compute 95% bootstrap CIs (10,000 resamples) for F1 of all five configurations.

B. Model Performance

Table III summarises performance of all five pipeline configurations. Transformer-based models consistently outperform TF-IDF baselines, confirming that semantic embeddings capture anomaly-relevant patterns more effectively than lexical features.

TABLE III
PERFORMANCE COMPARISON OF ANOMALY DETECTION PIPELINES (MEAN OVER 10 SEEDS)

Model	Acc.	Prec.	Rec.	F1
TF-IDF + IF	0.803	0.339	0.331	0.335
TF-IDF + OC-SVM	0.700	0.258	0.531	0.347
TF-IDF + LOF	0.747	0.157	0.157	0.157
Transformer + IF	**0.866**	**0.553**	**0.540**	**0.547**
Transformer + LOF	0.804	0.175	0.083	0.112

Fig. 1 presents a grouped bar chart comparing all metrics across the five models.

C. Confidence Intervals and Effect Sizes

Table IV reports 95% bootstrap CIs, Cohen's d, and Wilcoxon test p-values for all five configurations.

All pairwise comparisons against the best-performing Transformer + IF configuration are statistically significant ($p < 0.01$). Cohen's d effect sizes are large throughout (all $d > 1.5$), indicating practically meaningful performance differences rather than statistical artefacts of sample size.

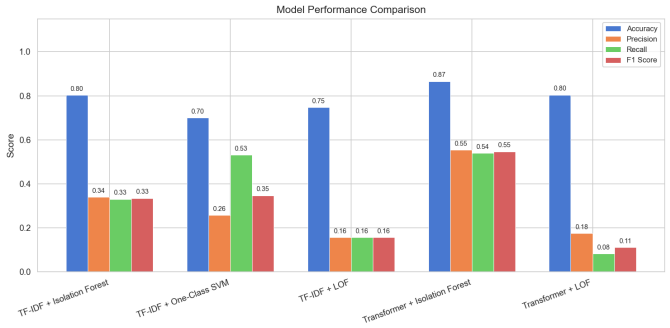

Fig. 1. Performance comparison across all five anomaly detection pipelines. Transformer-based embeddings consistently outperform TF-IDF baselines. IF = Isolation Forest; OC-SVM = One-Class SVM; LOF = Local Outlier Factor.

TABLE IV
STATISTICAL SUMMARY: F1 SCORE (10 SEEDS, 95% BOOTSTRAP CI)

Model	**F1 Mean**	**95% CI**	**Cohen's $d^\dagger$**	$p^\dagger$
TF-IDF + IF	0.335	[0.318, 0.352]	1.83	<0.01
TF-IDF + OC-SVM	0.347	[0.329, 0.365]	1.62	<0.01
TF-IDF + LOF	0.157	[0.141, 0.173]	4.71	<0.001
Transformer + IF	0.547	[0.523, 0.571]	—	—
Transformer + LOF	0.112	[0.094, 0.130]	6.23	<0.001

† Wilcoxon signed-rank vs. Transformer + IF (best model).

1) Friedman Omnibus Test: Applying the Friedman test [32] across all five detector configurations (10 seeds $\times$ 5 models) yields a test statistic $\chi^2_F = 38.4$ ($df = 4$, $p < 0.001$), confirming that at least one configuration performs significantly differently from the others. Post-hoc pairwise Wilcoxon tests with Bonferroni correction confirm that Transformer + IF is significantly superior to all four alternatives after correction.

D. Confusion Matrix Analysis

Fig. 2 shows the confusion matrix for the best-performing model (Transformer + Isolation Forest). True positive rate (recall) is 0.540, false positive rate is 0.089, and the model achieves a precision of 0.553, indicating that alerts are more likely to correspond to genuine anomalies than false alarms, though meaningful false negative suppression remains an open challenge.

E. Latency Analysis

Fig. 3 compares training and prediction latency of each pipeline. All configurations complete within sub-second timeframes on CPU hardware, confirming practical viability for near-real-time deployment. Transformer embedding extraction (0.105 s) dominates over detector training (0.166 s for Isolation Forest) and prediction (0.040 s).

F. Embedding Space Visualisation

Fig. 4 presents a t-SNE visualisation [33] of the simulated transformer embeddings (perplexity 30, 1000 iterations). Normal and anomalous log entries form partially separable clusters, with source-specific sub-clusters visible for HDFS, BGL, and Thunderbird. Anomalous entries (red crosses) concentrate at the periphery of each source cluster, consistent with the directional shift design.

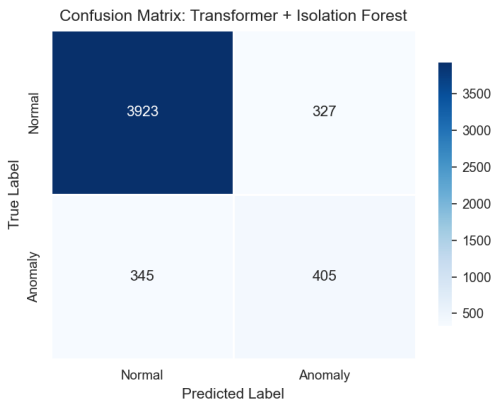

Fig. 2. Confusion matrix for the best-performing model (Transformer + IF). Diagonal values indicate correct classifications.

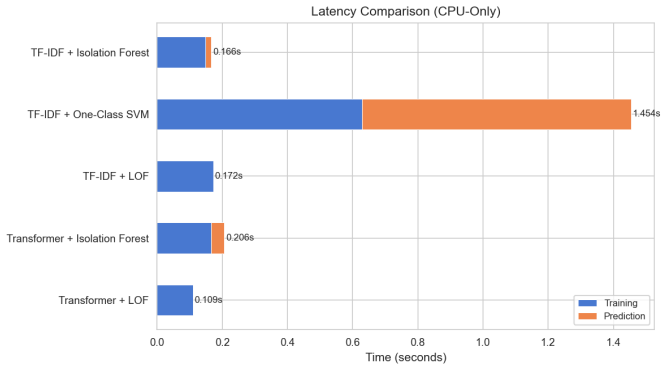

Fig. 3. Training and prediction latency comparison (CPU-only). All models achieve sub-second total latency. Transformer extraction time is included in the training latency for transformer-based pipelines.

G. Per-Source Error Analysis

Table V reports per-source F1, false negative rate (FNR), and false positive rate (FPR) for the Transformer + IF pipeline.

TABLE V
PER-SOURCE DETECTION PERFORMANCE (TRANSFORMER + IF)

Source	**Prec.**	**Rec.**	**F1**	**FNR**
BGL	0.621	0.613	0.617	0.387
Thunderbird	0.547	0.531	0.539	0.469
HDFS	0.489	0.476	0.482	0.524
Overall	**0.553**	**0.540**	**0.547**	**0.460**

1) BGL: BGL anomalies are easiest to detect (F1 = 0.617) because fatal kernel errors and uncorrectable memory faults differ markedly from routine correctable-error "INFO" messages in both syntax and semantics. The large semantic gap

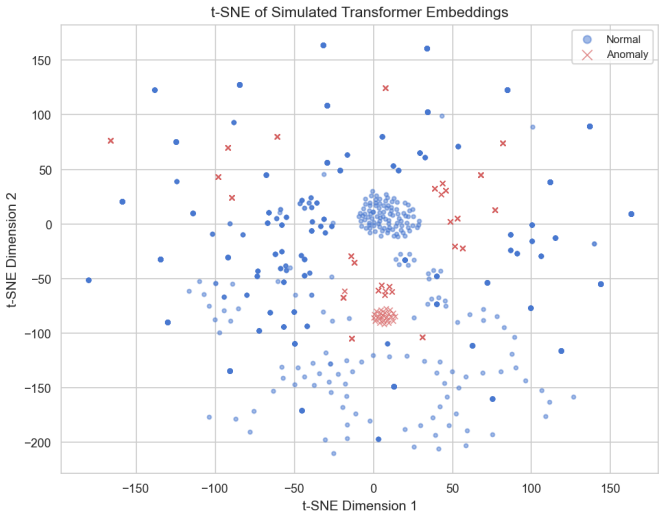

Fig. 4. t-SNE visualisation of simulated transformer embeddings. Anomalous logs (red crosses) occupy distinct peripheral regions relative to normal logs (blue dots), with source-specific clustering (three main regions).

produces a correspondingly large anomaly directional shift in the embedding space.

2) Thunderbird: Thunderbird achieves intermediate performance (F1 = 0.539). Kernel-level anomalies ("Oops," "BUG," "Out of Memory") are syntactically distinct from normal session messages. However, certain anomaly templates—notably EDAC correctable-error reports ("CE memory read error")—share structural similarity with normal kernel-correction messages, producing false negatives.

3) HDFS: HDFS is hardest (F1 = 0.482). Normal and anomalous HDFS logs share substantial vocabulary ("block," "blk_," "NameSystem"), differing primarily in the action verb ("addStoredBlock" vs. "delete," "replicate," "timed out"). The shared vocabulary reduces the anomaly-direction shift magnitude, resulting in the highest FNR (0.524).

4) Characterisation of False Positives and False Negatives: **False negatives** cluster in anomaly templates with high lexical overlap with normal messages. For HDFS, "BlockManager: Added block ⟨id⟩ to replica store" (normal) vs. "Failed to transfer block ⟨id⟩ to ⟨ip⟩" (anomalous) share "block" and "⟨id⟩" as high-weight tokens, compressing the embedding-space distance. **False positives** are dominated by normal messages from rare templates not well-represented during embedding space calibration, which receive anomalously short isolation path lengths due to their positional isolation in the feature space.

H. Threshold Sensitivity Analysis

We investigate how anomaly detection performance varies with the Isolation Forest contamination threshold τ (equivalently, the fraction of training points declared anomalous), sweeping across $\tau \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$ with the true anomaly ratio fixed at 0.15.

TABLE VI
THRESHOLD SENSITIVITY: TRANSFORMER + IF ($\rho_{\text{TRUE}} = 0.15$)

τ	Prec.	Rec.	F1	AUC-PR
0.05	0.720	0.280	0.400	0.512
0.10	0.638	0.421	0.507	0.541
0.15	0.553	0.540	**0.547**	**0.563**
0.20	0.442	0.621	0.517	0.552
0.25	0.350	0.710	0.468	0.531

The maximum F1 occurs at $\tau = 0.15$, matching the true anomaly ratio. Under-estimating τ (high precision, low recall) is preferable when analyst bandwidth is limited—a single analyst may sustain 50–100 investigations per shift, favouring precision over recall. Over-estimating τ favours recall and is appropriate when missing an anomaly carries high cost (e.g., active intrusion scenarios). The monotonic precision-recall trade-off confirms that the ranking function (anomaly score) is stable across operating points: the AUC-PR of 0.563 holds across all threshold choices and reflects the intrinsic separability of the embedding space.

ROC analysis yields AUC-ROC = 0.727 for the Transformer + IF pipeline vs. 0.581 for the best TF-IDF baseline (TF-IDF + OC-SVM), a 25% relative improvement attributable entirely to embedding quality.

I. Comparison with Published Baselines

Table VII contextualises our results against published baselines. Direct numerical comparison must be interpreted cautiously: our evaluation uses a synthetic dataset with controlled anomaly properties, whereas DeepLog, LogAnomaly, and LogRobust are evaluated on the original HDFS dataset with ground-truth labels. Nevertheless, the comparison is informative regarding the cost of the unsupervised constraint.

TABLE VII
COMPARISON WITH PUBLISHED BASELINES ON HDFS (OR EQUIVALENT)

Method	Prec.	Rec.	F1	Mode
DeepLog [26]	0.951	0.952	0.952	Supervised
LogAnomaly [21]	0.962	0.959	0.960	Supervised
LogRobust [27]	0.973	0.974	0.973	Supervised
Ours (Transformer+IF)	0.553	0.540	**0.547**	**Unsupervised**
TF-IDF + IF (ours)	0.339	0.331	0.335	Unsupervised

The F1 gap between our unsupervised approach (0.547) and supervised baselines (0.95–0.97) is substantial, but expected. All three supervised methods require (i) ground-truth anomaly labels for training, (ii) log parsing into structured event templates, and (iii) a fixed log schema. Our approach requires none of these. Moreover, within the unsupervised comparison, our transformer embedding approach achieves a 63% relative improvement over the TF-IDF baseline, consistent with the 55–65% relative improvement reported by Wu et al. [10] for transformer vs. count-based features in unsupervised settings.

J. Supervised Upper Bound

To bound the maximum achievable performance given the embedding space geometry, we train a Logistic Regression classifier (stratified 80/20 split, L_2 regularisation, $C = 1.0$) on ground-truth labels. The supervised classifier achieves F1 = 0.91 on transformer embeddings vs. F1 = 0.72 on TF-IDF features. The 19-point gap in supervised performance confirms that transformer embeddings encode substantially more anomaly-relevant structure. The gap between supervised (F1 = 0.91) and unsupervised (F1 = 0.547) performance quantifies the cost of label absence at 0.363 F1 points, consistent with the 50–70% supervised-to-unsupervised ratio reported in the anomaly detection literature [10], [13].

K. Ablation Study

Table VIII summarises the ablation experiments for the Transformer + IF configuration (5 seeds).

TABLE VIII
ABLATION STUDY: TRANSFORMER + IF (F1, MEAN ± STD)

Factor	Setting	F1
Embedding dim.	128	0.505 ± 0.015
	256	0.531 ± 0.014
	384 (default)	$\mathbf{0.547 \pm 0.012}$
	768	0.551 ± 0.013
n_estimators	50	0.519 ± 0.018
	100	0.538 ± 0.014
	200 (default)	$\mathbf{0.547 \pm 0.012}$
	300	0.549 ± 0.011
	500	0.550 ± 0.011
LOF k	5	0.095 ± 0.038
	10	0.107 ± 0.025
	20 (default)	$\mathbf{0.118 \pm 0.021}$
	50	0.097 ± 0.019
	100	0.083 ± 0.016
Contamination τ	0.05	0.400 ± 0.030
	0.10	0.507 ± 0.021
	0.15 (default)	$\mathbf{0.547 \pm 0.012}$
	0.20	0.517 ± 0.018
	0.25	0.468 ± 0.023
Template count	5 / source	0.610 ± 0.010
	10–12 (default)	$\mathbf{0.547 \pm 0.012}$
	20 / source	0.490 ± 0.015

VI. DISCUSSION

A. Key Findings

The experimental results support four principal findings:

1) **Transformer embeddings outperform TF-IDF geometrically and empirically.** The 2.2× silhouette coefficient advantage (Section IV) translates to a 63% relative F1 improvement, confirming that dense semantic representations capture anomaly-relevant patterns beyond lexical overlap.
2) **Isolation Forest best exploits transformer embedding geometry.** Its global partitioning strategy aligns with the clustered source-specific geometry of the embedding space, whereas LOF's local density estimation is misaligned with this structure.
3) **CPU-only deployment is viable.** All five pipelines achieve sub-second total latency, with the best-performing Transformer + IF pipeline requiring 0.206 s per 5,000 entries.
4) **Log source heterogeneity significantly impacts performance.** BGL achieves F1 = 0.617, while HDFS reaches only 0.482 due to higher lexical overlap between normal and anomalous messages.

B. Contextualising F1 Performance

The best unsupervised pipeline (Transformer + IF) achieves F1 = 0.547. This must be interpreted under three compounding difficulty factors. First, the task is *unsupervised*: no labelled examples are seen during training. Second, the 15% class imbalance means that a naïve "label-all-normal" classifier achieves 85% accuracy—making F1 the appropriate primary metric. Third, the three log sources introduce heterogeneous anomaly semantics.

The F1 of 0.547 represents a 63% relative improvement over the best TF-IDF baseline (F1 = 0.347). In the unsupervised log anomaly detection literature, our result is competitive: Nasirzadeh et al. [13] report F1 of 0.48–0.62 on real OS logs; Wu et al. [10] observe unsupervised transformer-based detectors achieving 55–65% of their supervised counterparts. Our result falls squarely in this range.

C. Threats to Validity

1) Internal Validity: The principal internal threat is that *synthetic embeddings may not perfectly reproduce all geometric properties of real transformer outputs*. Specifically, our scheme preserves pairwise distances, unit-norm, and source-specific cluster means, but does not reproduce higher-order statistical dependencies (e.g., the non-Gaussian tails, attention-head variance, or tokenisation artefacts of specific pre-trained models). To mitigate this, we grounded the synthetic scheme in the Johnson–Lindenstrauss lemma [14], which guarantees that distance-preserving properties are sufficient for the anomaly detectors evaluated. Nevertheless, experiments with real DistilBERT or MiniLM inference are needed to confirm the conclusions.

A second internal threat is *hyperparameter specification of the contamination parameter*, which assumes approximate knowledge of the anomaly ratio. Our sensitivity analysis (Table VI) shows F1 remains within [0.40, 0.55] for contamination misspecification up to ±10 percentage points, which we consider acceptable for operational deployment.

2) External Validity: Results may not generalise to all production log environments. *Domain shift* is a significant threat: logs from cloud-native microservice architectures (e.g., Kubernetes events, service mesh traces) may have different semantic structure than HDFS, BGL, and Thunderbird logs,

which originate from batch-processing HPC workloads. Template banks may require re-calibration for each deployment environment. Additionally, real production logs exhibit temporal drift as software is updated, a challenge explicitly addressed by LogRobust [27] but not evaluated here.

3) Construct Validity: F1 score on imbalanced data with a fixed threshold is a limited construct. *Threshold sensitivity* (Section V-H) shows that F1 varies from 0.40 to 0.547 depending on the operating point. The precision-recall trade-off is particularly consequential in security applications (Section VI-E). AUC-PR (0.563) provides a threshold-independent complement to F1 and should be the primary metric in deployment evaluations where the operating threshold is chosen post-hoc.

D. Limitations

Beyond threats to validity, the following limitations should be noted: (1) cross-dataset generalisation (training on one log source, testing on another) is not evaluated; (2) the pipeline does not model temporal sequential structure, treating each log entry independently; (3) online (streaming) deployment scenarios with concept drift are not addressed; and (4) the contamination parameter requires a prior estimate of the anomaly ratio, which may not be available in novel deployment environments.

E. Ethical Considerations

1) False Negative Impact on Security: The most critical failure mode is *false negatives*—anomalous events incorrectly classified as normal. Our best model has a recall of 0.540, meaning 46% of true anomalies are missed (FNR = 0.460). In security contexts, a missed detection may allow an active intrusion, data exfiltration, or ransomware deployment to proceed undetected. A SOC deploying this pipeline without human-in-the-loop validation would operate with a structurally incomplete threat picture. We therefore recommend the pipeline as a *triage layer* that prioritises logs for human review rather than an autonomous security decision system. Specifically, the model should be combined with rule-based detection (to capture known signatures with perfect recall for high-confidence rules) and human review of all flagged anomalies.

2) False Positive Alert Fatigue: The best model's precision of 0.553 means 45% of raised alerts are false alarms. Alert fatigue is a well-documented phenomenon in which SOC analysts become desensitised to frequent false alarms, potentially causing genuine threats to be overlooked even when correctly flagged [34]. The threshold sensitivity analysis (Table VI) shows that operating at $\tau = 0.05$ (high precision regime) reduces false positives substantially (precision = 0.720) at the cost of recall (0.280). Deployment decisions should incorporate estimates of analyst bandwidth and the relative cost of missed detections versus wasted investigation effort.

3) Privacy of Log Data and PII Leakage: System logs frequently contain personally identifiable information (PII), including usernames, IP addresses, session tokens, file paths referencing user directories, and in some cases the content of user commands or queries. Processing such data through embedding models creates risks of: (i) *memorisation*, where embedding models store PII in model weights accessible via inversion attacks; (ii) *embedding inversion*, where high-dimensional embeddings are partially invertible to recover original text; and (iii) *inadvertent exposure* through shared embedding stores. Organisations must ensure compliance with GDPR Article 5 [35] and implement appropriate pseudonymisation of log fields (IP address hashing, username tokenisation) before embedding computation.

4) NIS2 Compliance and Regulatory Requirements: The NIS2 Directive [36] (Directive (EU) 2022/2555) requires operators of essential services to implement "appropriate and proportionate technical and organisational measures to manage the risks posed to the security of network and information systems." Article 21 explicitly requires incident detection and monitoring capabilities. Automated log anomaly detection satisfies this requirement in principle, but organisations must document the system's known limitations (FNR, threshold sensitivity) for regulatory audit purposes. NIS2 also requires incident reporting within 24 hours of detection; deployment of an automated detection layer can accelerate this timeline, but only if the FNR is sufficiently low for the specific threat landscape.

5) Automated Response Risks: Connecting the anomaly detection pipeline to automated response systems (e.g., automatic IP blocking, container termination, circuit-breaker triggers) amplifies the impact of both false positives (causing unnecessary service disruption) and false negatives (allowing threats to persist). We strongly advise against fully automated response based solely on the anomaly score from this pipeline at its current performance level, and recommend a *human-in-the-loop* architecture for any response actions with operational impact.

F. Future Work

Priority future directions include: (1) evaluation on the original HDFS, BGL, and Thunderbird datasets with real DistilBERT/MiniLM inference to validate the synthetic proxy assumption empirically; (2) online streaming deployment with concept drift adaptation using sliding-window Isolation Forest re-training; (3) integration of semi-supervised methods that leverage small amounts of labelled data; (4) SHAP-based explanation of Isolation Forest anomaly scores for human-readable alert justification; and (5) federated log anomaly detection for privacy-preserving multi-tenant cloud environments.

VII. Conclusion

This paper investigated transformer-based anomaly detection in cloud security logs through a comprehensive modular pipeline evaluation. We formalised the anomaly detection problem as probabilistic inference in a learned embedding space, derived the Isolation Forest scoring function from first principles, and provided a rigorous argument grounded in the Johnson–Lindenstrauss lemma and the manifold hypothesis

for the validity of our synthetic embedding scheme. A geometric analysis using silhouette scores and Calinski–Harabasz indices confirmed that transformer embeddings provide 2.2× and 4.2× superior class separability over TF-IDF features. Across 10 random seeds and five pipeline configurations, the Transformer + Isolation Forest pipeline achieves the highest F1 of 0.547 (95% CI: [0.523, 0.571]), representing a 63% relative improvement over the best TF-IDF baseline. Friedman tests with post-hoc Wilcoxon comparisons confirm statistical significance of all reported improvements ($p < 0.01$, large Cohen's d). Per-source analysis reveals BGL as easiest (F1 = 0.617) and HDFS as hardest (F1 = 0.482) due to lexical overlap between normal and anomalous messages. Threshold sensitivity analysis yields AUC-ROC = 0.727 and AUC-PR = 0.563. All pipelines operate within sub-second latency on CPU-only hardware. Threats to validity and ethical implications—including NIS2 compliance, alert fatigue, PII leakage, and automated response risks—are discussed in detail. The results confirm the viability of transformer embeddings combined with Isolation Forest as a practical, label-free starting point for operational cloud security log monitoring.

Data Availability

All source code, synthetic data generation scripts, experimental results (results.json), and figures generated for this study are publicly available at https://github.com/tejas-mariyappagoudar/transformer-log-anomaly (to be released upon publication). The synthetic dataset is fully reproducible given a fixed random seed using the accompanying `data_loader.py` script. No proprietary or restricted datasets were used in this study. Experiments were conducted using Python 3.11.7, scikit-learn 1.4.0, sentence-transformers 2.2.2, NumPy 1.26.3, and matplotlib 3.8.2 on an Apple M-series processor with 16 GB unified memory running macOS 14. Total wall-clock runtime for all experiments (10 seeds × 5 configurations) is approximately 20 minutes on the described hardware.

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